

V01.10 – Mirror Physics IV

Mirror Physics IV**Causal Invariance and the Speed of Constraint Propagation**

Observerhood, Light Cones and the Physical Limits of Update

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Abstract

Mirror Theory treats observerhood as a recursive, viability-constrained process: a local system maintains a world-model, a self-model, reliability variables, memory governance and identity-continuity under perturbation. Previous papers in this sequence developed the mathematical language of constraint regimes, observer thresholds, identity-continuity, physical support conditions, constraint salience and thermodynamic maintenance. This paper turns to causal structure. It argues that persistent observerhood requires not merely information processing, but ordered and bounded causal update. In relativistic physics, the invariant speed c defines the boundary of causal influence and the structure of light cones. Mirror Theory does not modify this structure. Instead, it interprets c as the physical update bound within which memory, prediction, correction, self-modelling and identity-continuity can remain locally coherent. The central result is a Causal Support Theorem: any local observer that persists through time requires a physical regime supporting stable causal order, finite update propagation, local separability, memory preservation and non-instantaneous correction loops. The paper explicitly does not derive the numerical value of c , does not propose faster-than-light travel and does not alter relativity. Its contribution is interpretive and structural: it identifies why an invariant causal speed is not merely a feature of signal propagation, but a condition under which recursive observer-continuity can be physically supported.

Keywords: Mirror Theory; observerhood; causality; light cones; speed of light; constraint propagation; causal invariance; identity-continuity; thermodynamics; relativity; memory governance; physical support regimes; bounded update.

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1 Introduction

The speed of light is often introduced as a property of electromagnetic radiation. In modern relativity, however, c has a deeper status: it is the invariant speed appearing in spacetime structure, the limiting speed of causal influence in locally Lorentzian physics, and the parameter determining the light cone that separates timelike, lightlike and spacelike relations [22,25,26,28]. Light travels at c , but c is not merely about light.

Mirror Theory asks a different question. Instead of asking first whether c can be exceeded, modified or exploited, it asks why a universe capable of supporting persistent observers should require a stable causal update structure at all. A Mirror observer is not a disembodied subject. It is a local, viability-constrained system that must preserve memory, correct error, govern self-relevant updates and maintain identity-continuity across time [2–10]. Such processes require a physical substrate in which updates are ordered, bounded and locally meaningful. If causal influence were unordered, unbounded or globally instantaneous in a way that destroyed local separability, then memory, prediction, correction and self-model governance would lose their ordinary functional meaning.

This paper develops that point. It does not propose a new equation of motion. It does not modify special relativity or general relativity. It does not derive a new value of c . It instead supplies a Mirror-theoretic interpretation of causal invariance: the invariant speed c can be understood as the physical upper bound on causal constraint propagation in our regime, and such a bound is structurally necessary for persistent local observerhood.

The guiding claim is therefore:

Persistent observerhood requires bounded causal update. In our physical regime, c is the invariant update bound that structures the domain within which local memory, prediction, correction and identity can be maintained.

This is not a claim that observers create c , nor that c exists for the sake of observers. It is a compatibility and support claim: if recursive observers persist in a physical universe, then that universe must provide causal conditions under which local update is stable enough for recursive self-maintenance. Relativity provides one such structure through light cones and invariant causal ordering [22,25,26,28].

1.1 Position in the Mirror sequence

The preceding papers introduced the following sequence of concepts, within the Volume I architecture [1]:

- **Constraint calculus:** systems are described as constraint regimes $C = (S, T, I, V, \kappa)$, with states, transitions, invariants, viability functionals and bounded capacity [5].
- **Observer thresholds:** observerhood requires not merely prediction, but self-model reliability that is action-guiding, cost-sensitive and positively coupled to viability [6].
- **Memory governance:** identity is not memory accumulation; it is governed self-model continuity under perturbation [7].
- **Observerhood labs:** controlled simulations tested self-model reliability, channel-decomposed reliability, repair, recursive reliability and memory governance under perturbation [11–17].
- **Physical support:** observerhood requires causal order, memory substrates, thermodynamic throughput and error correction [8].
- **Constraint salience:** organisational structure may be measured by persistence-relevant information, causal coupling, reliability, error correction and cost [9].

- **Thermodynamic observerhood:** persistent observers are thermodynamically maintained constraint structures [10].

This paper adds causal invariance. It asks what kind of causal structure is required for such a system to persist.

2 Relativistic causal structure

In special relativity, spacetime intervals in flat Minkowski spacetime are often written as

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2, \quad (1)$$

with conventions varying by sign. The value c defines null directions, the boundary of light cones and the invariant relation between temporal and spatial separation [25,27,26]. Events inside the future light cone of an event can be causally influenced by it; events outside the light cone are spacelike separated and cannot be influenced by signals travelling at or below c .

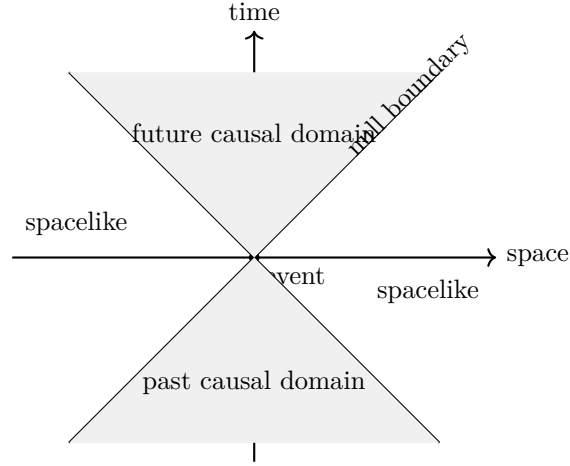


Figure 1: A light cone as a causal update boundary. Mirror Theory does not modify the light cone. It interprets the light cone as the physical boundary within which local update, memory, prediction and correction can remain causally coherent.

General relativity localises this structure: spacetime curvature changes the geometry, but local inertial frames remain Lorentzian and causal cones determine possible propagation of influence [20,28]. The important point for the present paper is not the detailed field equations, but the role of a stable causal cone in organising possible update.

2.1 From signal speed to update bound

The phrase “speed of light” can mislead. A more structural phrase is “invariant causal speed”. Within relativistic physics, c is the conversion factor and invariant speed that fixes causal structure [22,25,26,28]. In Mirror terms, it is the maximum speed at which physical constraint-updates can propagate through the regime in which local observers exist.

A constraint-update is not necessarily a conscious update or computational update. It is any physical influence that changes the state space available to a local system. If a photon reaches a detector, if a gravitational signal changes a measurable configuration, if a collision transmits momentum or if a thermal disturbance propagates through a medium, a local physical state is updated by causal influence. In relativistic physics, such updates are bounded by the causal structure determined by c .

3 Mirror observerhood as local causal update

Let L be a local system embedded in an environment E . In previous notation, L qualifies as a Mirror observer only if it maintains, across some horizon τ , enough of the following [5–10]:

- a world-model M_E ;
- a self-model M_L ;
- a reliability variable R_L ;
- memory governance G_L ;
- identity-continuity $I_L(t, t + \tau)$;
- a viability functional V_L ; and
- an action or update policy π_L .

These are not abstract decorations. Each requires ordered causal dependence. A memory must be physically encoded before it can be consulted. An error signal must arrive before it can be corrected. A self-model must be updated in response to evidence, not arbitrarily overwritten by every remote state of the universe. A repair action must occur after a perturbation is detected. Identity-continuity therefore presupposes causal ordering.

Definition 1 (Local observer process). *A local observer process over horizon τ is a tuple*

$$O_L^\tau = (M_E, M_L, R_L, G_L, V_L, \pi_L, I_L)_\tau \quad (2)$$

implemented by a physical subsystem L , such that the tuple is maintained by causal transitions within L and between L and its accessible environment over $[t, t + \tau]$.

Definition 2 (Causal accessibility). *For a local system L at event p , the accessible update domain $J^+(p)$ is the set of events that can be causally influenced by p . The accessible evidence domain $J^-(p)$ is the set of events that can causally influence p . In relativistic regimes these domains are determined by the light-cone structure.*

3.1 Why instantaneity is not automatically better

It is tempting to think that an observer would always benefit from faster influence. But unbounded instantaneity is not equivalent to better knowledge. If all states could influence all other states without causal ordering or locality, then a local system could lose the boundary conditions required for stable update. There would be no finite domain over which the system could distinguish evidence from noise, recent update from remote overwrite, or local self-state from global environmental entanglement.

Mirror Theory therefore does not treat finite causal speed as a limitation only. It is also an enabling constraint. A boundary on update allows a system to maintain local state, to perform correction before further perturbations arrive, and to model the environment from a located perspective.

Remark 1 (Constraint is not merely restriction). *A constraint can be enabling. The light cone restricts influence, but by doing so it also supports local causal order. Mirror Theory treats enabling constraint as central: a world with no stable update boundary would not obviously support bounded recursive observers.*

4 Causal support conditions

The following conditions are necessary for persistent Mirror observerhood in a physical regime.

Definition 3 (Causal support regime). *A physical regime P causally supports a local observer O_L^τ only if it provides:*

(C1) Ordered update:

causal transitions can be arranged into before/after relations sufficient for memory, prediction and correction.

(C2) Bounded propagation:

causal influence has a finite propagation structure, generating local domains of accessibility.

(C3) Local separability:

L can maintain a partial boundary from E , so that self-state and environmental state are not collapsed into an indistinguishable totality.

(C4) Memory preservation:

physical states can persist long enough to encode prior update.

(C5) Correction loops:

perturbations can be detected and acted upon before they destroy the relevant organisation.

(C6) Thermodynamic throughput:

energy and entropy flows allow continued maintenance of the observer process.

Theorem 1 (Causal Support Theorem). *If L is a persistent Mirror observer over horizon τ , then the physical regime implementing L must provide causal support conditions (C1)–(C6) over that horizon.*

Proof. A persistent Mirror observer maintains memory, self-model, reliability variables, governance and viability-directed action over time. Memory requires physically preserved prior states, giving (C4). Updating a model requires evidence or perturbation to be ordered relative to the update, giving (C1). Since L is local rather than identical with the entire universe, it requires an accessible domain and a distinction between self-state and environmental state, giving (C2) and (C3). Reliability tracking and repair require that perturbations can be detected and corrected before the relevant organisation is destroyed, giving (C5). Finally, maintaining non-equilibrium organisation, memory and correction requires physical work and entropy export, giving (C6). Therefore, persistent observerhood entails the causal support conditions. $\square$

The theorem is not meant to surprise relativists. It is meant to make explicit what Mirror Theory needs from physics. Observerhood is not implemented in a causally structureless void. It requires a causal architecture.

5 Finite propagation and the local update domain

Let $U_L(t)$ denote the update state of L at time t . Let ∂L denote the effective boundary through which information, energy and perturbations enter or leave L . Let $v_{\max}$ be the maximum speed of causal influence in the relevant regime. In relativistic physics, locally,

$$v_{\max} = c. \quad (3)$$

For a finite observer with characteristic size ℓ_L , there is a minimum internal update time scale on the order of

$$\tau_{\text{cross}} \geq \frac{\ell_L}{c}, \quad (4)$$

ignoring medium-dependent delays, internal processing and thermodynamic costs. This is not a new relativistic equation. It is the ordinary causal crossing time interpreted as an observerhood

constraint: no extended physical observer can update all of itself faster than causal influence can traverse its relevant extent.

Lemma 1 (Finite Update Lemma). *For any extended physical observer L with nonzero characteristic extent ℓ_L , globally coherent internal update cannot be physically completed on timescales shorter than the causal crossing time ℓ_L/c , except by relying on pre-established correlations or local approximations rather than new causal influence.*

Proof. A globally coherent internal update requires that the relevant parts of L either exchange causal influence or were already coordinated by prior physical structure. New causal influence cannot propagate faster than c in a relativistic regime. Therefore, new global coordination across extent ℓ_L requires at least the causal crossing time. Pre-established correlations may reduce the need for new signalling, but do not transmit new causal influence faster than c . $\square$

5.1 Prediction depends on delay

Prediction matters because causal influence is not instantaneous. If every effect were immediately available everywhere without ordered propagation, prediction would lose much of its functional role. In a bounded causal regime, however, a local system must act under partial information. It must infer future states before all consequences are locally available. This is why modelling, compression and reliability tracking become viability-relevant [2,5,6].

Proposition 1 (Bounded propagation makes modelling useful). *In a regime where local systems receive only bounded and delayed causal updates, internal models can improve viability by estimating unavailable future or remote states. Therefore, bounded propagation is one condition under which modelling can become viability-relevant.*

6 Light cones as observerhood domains

Mirror Theory interprets the light cone not merely as a boundary in spacetime diagrams, but as the update domain of local observerhood:

- the past light cone bounds what can have informed the observer;
- the future light cone bounds what the observer can affect; and
- the spacelike region marks what cannot yet be causally integrated into the observer's state.

This matters for self-models. A local observer cannot model reality from nowhere. Its model is built from its causal history. Objectivity, in this view, is not a view outside all light cones; it is convergence across many local causal histories under repeated correction.

7 Causal invariance and the meaning of c

A central lesson of relativity is that observers may disagree about simultaneity while agreeing on invariant structure [22,25,26,28]. Mirror Theory uses the term “observer” differently from inertial observer in physics: it refers to a self-maintaining recursive system. Nevertheless, the compatibility is suggestive. Physical observers are local. They access the world through causal structure. They do not possess a global now.

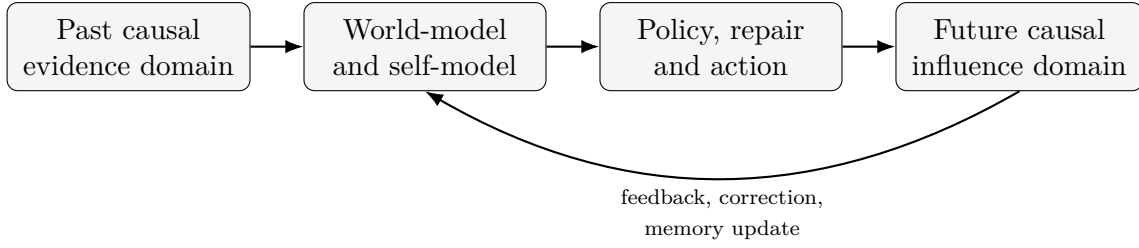


Figure 2: The Mirror update loop as a causal process. Evidence must be available in the past causal domain, incorporated into governed models, used by policy and then expressed through future causal influence.

Definition 4 (Causal invariance). *Causal invariance is the preservation of causal structure across admissible descriptions of a physical regime. In special relativity, Lorentz transformations preserve the spacetime interval and light-cone structure.*

Mirror Theory adds an interpretive claim:

The invariance of causal structure is what allows local observers to disagree perspectivaly while still inhabiting a shared constraint regime.

If causal structure were not invariant in this sense, the possibility of intersubjective objectivity would be weakened. Different observers could not reliably reconcile their local models into a shared account of events.

7.1 The speed of constraint propagation

We can now state the disciplined Mirror interpretation:

In our physical regime, c is the invariant upper bound on causal constraint propagation. It defines the maximum speed at which new physical update can enter the local modelling and correction domain of an observer.

This statement is not a new law. It is a reinterpretation of relativistic causal structure in observerhood language. The phrase “constraint propagation” refers to physical update of possible states, not to an additional substance or field.

Corollary 1 (Causal update bound). *If a local observer L is implemented in a relativistic physical regime, then any new external influence on L ’s memory, self-model, reliability variables or action policy must enter through L ’s past light cone and cannot propagate faster than c .*

Proof. Memory, self-model, reliability variables and policy are physical states of L . New external influence on such states is a causal influence. In a relativistic regime, causal influence lies within or on the light cone and is bounded by c . $\square$

8 Why bounded causal propagation supports identity

Identity-continuity, as formalised in prior Mirror work, is not mere storage. It is governed self-update. Candidate memories are committed, quarantined, audited or rejected depending on reliability and viability relevance [7,10]. Such governance requires that updates arrive in an order and at a rate compatible with integration.

Let G_L be a memory governance function. Let m_t be a candidate self-relevant update arriving at L . A simplified governance structure is

$$G_L(m_t) \in \{\text{commit, quarantine, audit, reject}\}. \quad (5)$$

For G_L to operate, m_t must be locally available, comparable to existing identity structure, and integrated or rejected before its effects destabilise the system. This assumes finite update rates. If arbitrary self-relevant updates could globally overwrite L without boundary or delay, governance would be bypassed.

Theorem 2 (Governed Identity Requires Bounded Update). *A local system cannot maintain governed identity-continuity under arbitrary self-relevant perturbation unless self-relevant updates are bounded by an integration regime that permits evaluation, correction or rejection.*

Proof. Governed identity-continuity requires that candidate self-relevant updates be evaluated relative to existing self-model, memory and reliability structure. Evaluation requires a temporal relation between prior identity state, candidate update and integration outcome. If updates are not bounded by any integration regime, then candidate self-relevant perturbations can overwrite or destabilise the identity state without evaluation. In that case governance fails. Therefore bounded update is necessary for governed identity-continuity under perturbation. $\square$

This theorem does not say the bound must numerically equal c in every possible universe. It says that some bounded update structure is necessary. In our universe, relativistic causal structure provides the fundamental upper bound.

9 Objectivity as causal convergence

Mirror Theory does not require a view from nowhere. Instead, objectivity is model convergence across local causal perspectives under correction. This is compatible with relativity: observers may occupy different frames and causal histories, but invariant relations permit translation and reconciliation [22,25,28].

Let M_i and M_j be models maintained by local observers i and j . Let T_{ij} be an admissible transformation between their descriptions. Objectivity increases when

$$d(T_{ij}(M_i), M_j) \rightarrow 0 \quad (6)$$

under repeated evidence exchange and correction, where d is a model-distance measure appropriate to the domain.

Causal invariance matters because it makes such transformation meaningful. Without stable causal structure, observers would not be able to reconcile local models into shared invariants.

10 No faster-than-light conclusion

This paper is intentionally conservative. It does not infer any of the following:

- that c can be exceeded by engineering;
- that Mirror Theory modifies Lorentz invariance;
- that observerhood changes spacetime dynamics;
- that consciousness has a causal field;
- that constraint salience couples to gravity or electromagnetism; or
- that faster-than-light communication or travel is possible.

The paper instead argues that c , as the invariant causal speed, has observerhood significance: it structures the update domain within which persistent local observers can exist.

10.1 What would be required for new physics?

A genuine physical extension would require much more. At minimum it would need:

1. a precise physical quantity, not merely an organisational metaphor;
2. compatibility with Lorentz symmetry or a clear derivation of symmetry breaking;
3. reduction to tested relativity in known regimes;
4. a Lagrangian, field equation, statistical mechanical derivation or other accepted mathematical foundation;
5. at least one novel prediction distinguishable from standard physics; and
6. an empirical test capable of falsifying the new claim.

Until those conditions are met, Mirror Physics remains an interpretive and structural programme, not a replacement for established physics. This discipline is continuous with the compatibility requirements stated in Mirror Physics I and the strong-coupling discipline stated in Mirror Physics II [8,9].

11 Engineering implications

Although this paper does not imply faster-than-light travel, it suggests a disciplined engineering philosophy. Advanced technology is not primarily the violation of constraints; it is the discovery of exploitable structure within constraints. If c is the update bound of our physical regime, then engineering must work with causal delay rather than wish it away.

For interstellar engineering, this implies emphasis on:

- autonomous local decision-making, because remote control is causally delayed;
- robust memory and error correction, because repair cannot rely on immediate external intervention;
- distributed probes and redundant systems;
- predictive modelling under delay;
- efficient use of local resources; and
- communication protocols designed around light-cone separation.

This is a practical consequence of Mirror Physics IV. The path to deep-space systems is not to ignore c , but to build agents whose local observerhood is strong enough to survive under causal separation.

12 Relation to prior physical frameworks

The present account is compatible with several established lines of thought:

- **Relativity:** c defines invariant causal structure, not merely light propagation [22,25–28].
- **Information and thermodynamics:** memory and computation have physical costs [19,24].
- **Cybernetics:** control systems require feedback, delay management and requisite variety [18,29].
- **Active inference and predictive processing:** systems act under uncertainty and partial information [21,23].

Mirror Theory differs by asking when these ingredients jointly support persistent observerhood: not mere response, not mere prediction, but governed recursive self-maintenance across causal time.

13 Conclusion

Mirror Physics IV returns to the speed of light, but not by attempting to break it. The invariant speed c is interpreted as the causal update bound of our physical regime: the boundary governing how new physical influence can enter the local domains in which memory, prediction, correction and identity-continuity occur.

The central claim is structural. Persistent observers require ordered and bounded causal update. In relativistic physics, light cones supply this architecture. The speed of light is therefore not merely an obstacle to interstellar travel. It is part of the causal structure that makes local observerhood possible at all.

This does not derive the numerical value of c , modify relativity or license faster-than-light speculation. It does something more modest and, within the Mirror Programme, more important: it explains why causal invariance belongs inside a theory of observerhood. A world capable of producing persistent observers must not only contain matter and energy. It must support local, temporally ordered, bounded update. In our world, c is the invariant limit of that update.

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